

Algebra II with Trigonometry

Chapter 7.4 & 7.5

Study guide

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1. Short Summary

This document is a study guide for solving trigonometric equations typically covered in Algebra 2 Trigonometry/Honors, specifically focusing on sections 7.4 and 7.5. It explains strategies for solving basic trigonometric equations as well as more advanced types that require techniques like factoring, using identities, solving equations with multiple angles, and applying sum/product formulas. The guide provides worked examples and practice equations for students to solve.

2. Core Definitions, Theorems, and Formulas

Definitions

- **Trigonometric Equation:** An equation containing trigonometric functions of angles.
- **Basic Trigonometric Equation:** An equation of the form $T(\theta) = c$, where T is any trig function and c is a constant.

Theorems & Properties

- **Zero-Product Property:** If $AB = 0$, then $A = 0$ or $B = 0$.
- **Pythagorean Identity:** $\sin^2 \theta + \cos^2 \theta = 1$.

Key Formulas and Approaches

- **Solving Basic Trigonometric Equations:**

1. Find solutions in one period. 2. Express all solutions by adding integer multiples of the period to the fundamental solutions.

- **Multiple Angles:**

- When solving for functions of multiple angles, first solve for the multiple, then divide to get solutions for the angle itself.

- **Common Trigonometric Identities:**

- **Double-Angle:** $\sin(2\theta) = 2 \sin \theta \cos \theta$, $\cos(2\theta) = \cos^2 \theta - \sin^2 \theta$
 - **Sum and Difference:** $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$
 - **Product-to-Sum, Sum-to-Product:** Conversions between sums and products.
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3. Study Guide: Step-by-Step Explanation of Important Ideas

I. Solving Basic Trigonometric Equations

Example Structure:

Given: $\sin \theta = c$ **Step 1:** Find all solutions for θ in one period (e.g., $0 \leq \theta < 2\pi$ for sine and cosine). **Step 2:** Generalize by adding integer multiples of the period:

- For sine/cosine: $\theta = \theta_0 + 2\pi n$
 - For tangent: $\theta = \theta_0 + \pi n$
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II. Solving by Factoring

When the equation can be written in a polynomial-like form, move all terms to one side, factor, and use the Zero-Product Property.

Example: $2 \sin^2 \theta - 3 \sin \theta + 1 = 0$ 1. Factor: $(2 \sin \theta - 1)(\sin \theta - 1) = 0$
2. Use Zero-Product:

- $2 \sin \theta - 1 = 0 \implies \sin \theta = \frac{1}{2}$

- $\sin \theta - 1 = 0 \implies \sin \theta = 1$

3. Solve for θ .

III. Using Identities and Multiple Angles

- Use trig identities to rewrite equations in a solvable form.
 - For functions like $\sin(2\theta)$, use double-angle or other relevant formulas.
 - When equations mix multiple angles, introduce substitution or reduce using identities.
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IV. Graphical and Algebraic Solutions

- For equations difficult to solve algebraically (e.g., $\sin x = x/3$), use a graphing device.
 - Round graphical solutions to two decimal places.
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V. Expressing Solutions

- **General solutions:** Express using n (an integer) to denote all possible answers.
 - **Specific interval:** Give solutions in $[0, 2\pi)$ or other indicated intervals.
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4. Practice Problems with Answers

1. Solve $\cos \theta = -\frac{1}{2}$ for all θ .

Answer: $\theta = \frac{2\pi}{3} + 2\pi n$ and $\theta = \frac{4\pi}{3} + 2\pi n$ for all integers n .

2. Solve $2 \sin^2 \theta - 1 = 0$ for $0 \leq \theta < 2\pi$.

Solution: $2 \sin^2 \theta = 1 \implies \sin^2 \theta = \frac{1}{2} \implies \sin \theta = \pm \frac{\sqrt{2}}{2}$ So,
 $\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$.

3. Solve $\tan(2\theta) = 1$ for $0 \leq \theta < 2\pi$.

Solution: $2\theta = \arctan(1) + \pi n \implies 2\theta = \frac{\pi}{4} + \pi n \implies \theta = \frac{\pi}{8} + \frac{\pi}{2}n$, for $n = 0, 1, 2, 3$:

- $n = 0 : \frac{\pi}{8}$
 - $n = 1 : \frac{5\pi}{8}$
 - $n = 2 : \frac{9\pi}{8}$
 - $n = 3 : \frac{13\pi}{8}$
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4. Solve $\sin \theta(2 \cos \theta - 1) = 0$ in $[0, 2\pi)$.

Solution:

- $\sin \theta = 0 \implies \theta = 0, \pi$
 - $2 \cos \theta - 1 = 0 \implies \cos \theta = \frac{1}{2} \implies \theta = \frac{\pi}{3}, \frac{5\pi}{3}$
 - **Final answers:** $0, \pi, \frac{\pi}{3}, \frac{5\pi}{3}$
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5. Use factoring to solve $4 \cos^2 \theta - 3 = 0$ for all θ .

Solution: $4 \cos^2 \theta = 3 \implies \cos^2 \theta = \frac{3}{4} \implies \cos \theta = \pm \frac{\sqrt{3}}{2}$ So
 $\theta = \frac{\pi}{6} + 2\pi n, \frac{11\pi}{6} + 2\pi n, \frac{5\pi}{6} + 2\pi n, \frac{7\pi}{6} + 2\pi n$.

5. Assumptions or Ambiguities Resolved

- **Symbols:** OCR or typesetting errors (for example, incomplete equations) were interpreted using standard trigonometric problem forms.
- **Periods:** Assumed standard periods for sine/cosine (2π), tangent (π), unless otherwise specified.
- **Interval:** When the interval wasn't specified, both general solutions and solutions in $[0, 2\pi)$ were presented.
- **Graphing Solutions:** For problems requiring graphing (like $\sin x = x/3$), students are expected to use a calculator or graphing tool; a general “round to two decimals” rule is applied.

- **Formulas:** Typical trig identities (such as double/sum/difference formulas) were referenced by standard forms as the document mentions their use but doesn't list them explicitly.
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End of Study Guide